

Joel Rogers

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PROFESSIONAL SUMMARY

Mechatronics Engineer at Swarm Intelligence and penultimate year **Mechatronics Engineering student** at the University of Canterbury, specialising in **GNSS/GPS, microcontroller programming** and **wireless data communication (Radio, WiFi and 4G)**. Experience designing and integrating **avionics** into real world aerospace platforms. Industry experience delivering **embedded programming (C++)**, **PCB design** and **system automation** in environments where reliability and performance are critical.

EXPERIENCE

Professional

Mechatronics Engineer, Swarm Intelligence

Christchurch, Canterbury, March 2026-Present

- My most involved project has consisted of **server construction, server coding, ESP electronic coding, GNSS tracking, hardware integration** and **wireless data communication**.
- PCB design (KiCAD)** for microcontroller and daughter boards in real world mobile systems.
- Designing enclosures and mounting systems for electronics. (**SolidWorks, 3D printing**)
- Pitching my projects possibilities to potential clients in board room meetings, creating briefs and a demonstration video.

Electronics/Robotics Engineering Intern, Swarm Intelligence

Christchurch, Canterbury, October 2025-March 2026

- Design and implementation** of automation and embedded control systems deployed in high-performance engineering and manufacturing environments (including projects with **Pratt & Whitney** and **Fulton Hogan**).
- A broad range of projects from **wireless microcontroller programming** to **automation integration**.

Tiler, Syrus Imperial Tiling Company LTD

Christchurch, Canterbury, December 2023- November 2025

- This experience allows me to combine a tradesman's **hands-on work ethic** with a strong analytical approach to bridge the gap between design and **real-world implementation**.

Technical Projects

UAV (Personal)

- Designed a **flight controller (KiCAD)** with an integrated GPS and IMU module for autonomous flight and FPV camera control.
- Designed, modelled (**SolidWorks**), manufactured and tested a flying wing UAV platform.
- Integrated onboard **avionics**, including an inertial measurement unit, flight controller, GPS module, ESC motor controller and power regulating/distributing system.
- Integrated long range **RF** communication and **GPS** telemetry for **command and position tracking**.

Line Following Robot (University)

- Designed (**KiCAD**) and assembled a fully working main **microcontroller PCB** and smaller sensor PCBs.
- Broke the record** for the annual University of Canterbury line following robot competition, with our fastest time being **5.81s**.
- Designed, 3D modelled (**Fusion 360/SolidWorks**), **3D printed** and tested numerous gearing systems, motor mounts and chassis.
- Programmed the robot in **Arduino IDE** and integrated an ESP8266 for **wireless communication** and PID value changes over **WIFI**.

STM32 Step Counter (University)

- Programmed (**STM32CubeIDE**) an STM32 walking step counting system using **interrupt** driven software.
- Steps were detected through reading an **IMU** and implementing threshold based counting.
- Used **GIT** for version control.

SKILLS

Programming:

- Python
- C, C++, C#, Arduino
- HTML
- JavaScript
- GIT

Embedded & Electronics:

- KiCAD
- Soldering and electronic assembly
- Sensor integration
- AutoCAD Electrical
- CX Programmer
- STM32CubeIDE
- Arduino IDE

Digital

Communication:

- WiFi
- Ethernet
- SMS
- GNSS tracking
- MQTT
- Modbus
- RF

Mechanical:

- SolidWorks
- 3D printing
- Fusion 360
- Lathe
- Mill

Soft Skills:

- Leadership** from my time leading group projects at university, coaching hockey and taking charge of the bar at The Commodore hotel.
- Relationship building** from interacting with customers at The Commodore Hotel and meeting new coworkers at Swarm Intelligence.

Further Experience

- Workplace First Aider** Certified (Valid until February 2028)

Professional:

- Bartender**, The Commodore Hotel (November 2022 – February 2025)
- Hockey Coach**, St Andrews College (April 2023-August 2025)
- Representative Hockey Coach**, Canterbury Hockey (August 2025 – September 2025)
- Representative Hockey Player**, Canterbury Hockey (Canterbury U18: April 2021-August 2023)

Technical Projects:

- Designed and trained** multiple **AI lecture bots** to reteach me University content when revising for exams. (Personal)
- Analysed and tweaked in code a **machine learning model** for image edge detection. (University)
- Microcontroller LED **"Tic Tac Toe" game**. (University)
- CNC** modular pick and place. (University)
- PLC elevator** control system. (University)
- Aluminium Structure (University)
- PID Robot Cart (University)
- Arduino based **bike thief alarm**. (Personal)
- Arduino based **garage door sensor**. (Personal)
- Numerous Projects through my time at **Swarm Intelligence** (Swarm Intelligence)

EDUCATION

Bachelor of Engineering (Honours) in Mechatronics

Christchurch, University of Canterbury,

Expected graduation: 2027

References can be produced on request